

A comprehensive intrusion detection method for the internet of vehicles based on federated learning architecture

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ABSTRACT

Cybersecurity breaches within the Internet of Vehicles (IoV) have been increasingly reported annually with the proliferation of intelligent connected vehicles. Two primary obstacles are faced by current intrusion detection systems: substantial computational demands and stringent data privacy regulations, complicating both efficient deployment and the safeguarding of data privacy. Consequently, there is a pressing need for intrusion detection solutions that are both efficient and considerate of privacy concerns. This paper introduces FED-IoV, an innovative intrusion detection method tailored for the IoV, leveraging a federated learning architecture. FED-IoV aims to collaboratively perform detection tasks across distributed edge devices, thereby minimizing data privacy risks. Vehicular communication traffic data is transformed into images, and a bespoke, efficient model, MobileNet-Tiny, is employed for feature extraction, rendering FED-IoV capable of achieving high detection accuracy whilst being viable for deployment on devices with limited resources. Through evaluation against the authoritative datasets CAN-Intrusion and CICIDS2017, exceptional accuracy rates of 98.51 % and 97.74 %, respectively, were demonstrated by FED-IoV within a federated learning context, and excellent detection capabilities on imbalanced datasets were also shown. Moreover, a prediction latency of under 10 milliseconds per sample was maintained on devices with limited computational power, such as the Raspberry Pi 4 8GB, showcasing significantly better accuracy and real-time performance relative to existing approaches. The successful deployment of FED-IoV ushers in a novel, privacy-preserving, and efficient intrusion detection solution for IoV security.

1. Introduction

In the era of 5 G, where everything is interconnected, information and communication technology are rapidly merging with the automotive industry. This convergence makes smart cars and autonomous driving the benchmarks for industry research and development. However, behind the booming rise of the smart vehicle industry, the Internet of Vehicles (IoV), as a crucial asset of transportation's critical information infrastructure, is gradually becoming a target for hackers. The IoV is an open and integrated network system established through the interconnection of vehicles, wireless networks, and other units (Yang et al., 2021); it serves as a crucial mode of communication for modern intelligent automobiles. According to the "2023 Global Automotive Industry Cybersecurity Report" published by Upstream, losses amounting to 505 billion USD have been suffered by the global automotive industry over

the past five years due to cyberattacks. Furthermore, among all automotive cybersecurity incidents, >70 % are accounted for by events triggered by remote cyberattacks.

In 2022, a high-risk security vulnerability in the SiriusXM Internet of Vehicles (IoV) service platform was exposed. This vulnerability allowed hackers to attack vehicles from several manufacturers, including Honda, Nissan, and Infiniti. Researchers indicated that with just the vehicle identification number (VIN), the flaw could be exploited to unlock, start, locate, and honk the horn of the vehicles in an unauthorized manner.

The increasing number of cyberattacks targeting the IoV has raised concerns about its security, as such attacks can lead to severe consequences such as vehicle inoperability or traffic accidents. In response to this issue, intrusion detection models have become a key measure to ensure the security of IoV network communications. Intrusion detection methods identify signs of network attacks by collecting and analyzing

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the flow of communication data within the IoV, facilitating the classification of intrusion attacks and normal communications (Lampe and Meng, 2023). This security measure is instrumental in maintaining the stability and safety of the IoV.

Deep learning has been deemed to hold significant promise for intrusion detection (Laghrissi et al., 2021; Singh et al., 2021; Nguyen and Kim, 2020; Ma et al., 2017; Lotfollahi et al., 2020; Shone et al., 2017). These methods start by collecting adequate network traffic data packets, then analyze these packets to extract feature vectors, which serve as inputs for the classification model. By training deep learning models, an intrusion detection classifier is constructed, significantly improving detection accuracy compared to traditional signature-based and machine learning-based methods.

For instance, Laghrissi et al. (Laghrissi et al., 2021) proposed an intrusion detection system based on Long Short-Term Memory (LSTM), utilizing Principal Component Analysis (PCA) and mutual information for feature extraction in the preprocessing stage. The top three features identified by PCA were input into the LSTM model for training. Singh et al. (Singh et al., 2021), focusing on the dimensional features and time series characteristics of the data, simplified data features using PCA and trained a Gated Recurrent Unit (GRU) detection model using transfer learning on the simplified features. Nguyen et al. (Nguyen and Kim, 2020) introduced a Genetic Convolutional Neural Network model, initially employing a genetic algorithm that combines the KNN fitness function and fuzzy C-means clustering for feature selection, resulting in an improved feature subset, which is then input into a Convolutional Neural Network model for classification. Ma et al. (Ma et al., 2017), leveraging the strong generalization capability of the Extreme Learning Machine (ELM), achieved good detection results. Lotfollahi et al. (Lotfollahi et al., 2020) used stacked autoencoders and convolutional neural networks to classify encrypted traffic. Shone et al. (Shone et al., 2017) designed a stacked asymmetric deep autoencoder, using a random forest algorithm as the classifier to identify network attack categories.

Despite the unprecedented accuracy achieved by deep learning methods in intrusion detection, two fundamental challenges remain for intrusion detection in the IoV scenario. Intrusion detection for IoV demands low computational power consumption and real-time performance. However, most deep-learning models require extensive GPU computations, and they have not been tested on low-power platforms, restricting their deployment on in-vehicle edge devices. Centralized, deep-learning-based intrusion detection methods inherently rely on sufficient datasets. However, due to data protection regulations, countries have introduced the General Data Protection Regulation (GDPR) since 2017 (Regulation (EU), 2018), imposing strict restrictions on data aggregation and use. Smart vehicle owner information contains crucial data; thus, aggregating communication traffic data raises privacy concerns, potentially leading to data silos.

To effectively address the challenges posed by computational resource limitations in intrusion detection within the Internet of Vehicles (IoV) environment, a viable strategy is the combination of graphical representation of traffic data with compression techniques for deep neural network models. Specifically, Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in the field of computer vision, particularly in automatic feature extraction and complex pattern recognition. Transforming communication traffic data into image form is not merely a means to intuitively display data features but also reduces the demand for traditional data analysis and reverse engineering techniques. Furthermore, by implementing pruning techniques targeted at CNN models, the model size can be significantly reduced, thus lowering its computational resource requirements. This process eliminates redundant or insignificant parameters in the model, resulting in a more streamlined and efficient network capable of operating on resource-constrained in-vehicle edge devices, ensuring real-time performance while reducing resource consumption.

To effectively tackle the privacy protection issues in intrusion detection within the IoV environment, a potential method is federated

learning. This concept, initially proposed by Google in 2016 (McMahan et al., (McMahan et al., 2017)), aims to construct a global model by aggregating model parameter updates from various clients without sharing the clients' raw data. The core idea is to utilize privacy-preserving mechanisms, such as differential privacy or secure multi-party computation, to securely aggregate model updates from decentralized data sources, thereby producing a unified model that is superior in performance and does not require sensitive data to be exposed. The introduction of Federated Learning not only helps protect data privacy and prevent data leakage risks but also facilitates knowledge sharing among various local models through remote collaborative training, achieving detection performance close to centralized training approaches. Federated Learning has rapidly developed in security domains such as malware detection (Zhang et al., 2023), healthcare (Lakhan et al., 2023), privacy protection (Song et al., 2024), Internet of things (Wang et al., 2024), and mobile edge computing (Wang et al., 2022).

In this paper, FED-IoV, a comprehensive intrusion detection architecture for the Internet of Vehicles environment, is introduced. Addressing the challenges posed by the implementation of most current deep learning-based intrusion detection methods on platforms with limited computational resources, as well as the unique data privacy concerns inherent in the IoV, a federated learning strategy is adopted to ensure the privacy of user data during the aggregation process. Under the federated learning framework, vehicular communication traffic data are transformed into three-channel color images, and a pruned, optimized lightweight MobileNet-Tiny neural network model is designed. This approach not only ensures the accuracy of detection but also reduces detection latency. Evaluation experiments were conducted on Raspberry Pi devices using two representative real-world vehicular network datasets, simulating performance under real-world IoV conditions. Specifically, the principal contributions of this paper are:

- An innovative preprocessing technique has been introduced, transforming vehicular communication traffic data into images to leverage the advantages of Convolutional Neural Networks (CNNs) in image processing. Concurrently, an optimized and pruned MobileNet-Tiny model has been designed, achieving high accuracy and real-time detection capabilities while adhering to the computational resource constraints of in-vehicle systems.
- To protect vehicle owners' privacy and address data-sharing issues associated with traditional deep learning methods, this paper proposes FED-IoV, a novel intrusion detection framework for the Internet of Vehicles (IoV). This framework employs the principles of federated learning, allowing multiple independent and identical deep training models to construct a comprehensive detection model. Furthermore, FED-IoV enables each client to process data independently, effectively mitigating the risk of data leaks.
- The model has been theoretically proposed and experimentally validated on low-cost, computationally limited devices, namely Raspberry Pi, using real vehicular network datasets to evaluate the architecture's actual performance and feasibility. This validation underscores FED-IoV's practicality and provides robust support for real IoV scenarios.

The remainder of the paper is organized as follows: Section II primarily reviews the latest research findings in IoV intrusion detection and federated learning. In Section III, we focus on introducing the preprocessing method and model, with a detailed explanation of the FED-IoV scheme. Section IV assesses the performance of the proposed scheme through extensive experiments. Finally, Section V concludes the paper.

2. Background and related work

2.1. Internet of vehicles intrusion detection

With the rapid development of artificial intelligence technology, especially in the area of deep learning, methods for intrusion detection in the Internet of Vehicles (IoV) based on deep learning have gradually become a focal point of research. Relying on their exceptional ability to extract data features and efficiency in recognizing complex patterns, these methods have been provided a robust approach for detecting and defending against potential security threats within IoV systems. In this context, a variety of security threats in the IoV setting, such as malware, DDoS attacks, false data injections, and privacy leaks, have been the focus of researchers employing deep learning techniques to find solutions (Lampe and Meng, 2023). Xie et al. (Xie et al., 2021) analyzed security threats to CAN networks and proposed an intrusion detection technique enhancement employing GAN models. By using the vehicular CAN communication matrix as input, all CAN messages from the same sender are aggregated into blocks of data, which then serve as the input for the detection model. Through a series of experimental validations, it was proved that this novel detection technique is effective at identifying and defending against a variety of malicious activities, including DoS, message injection attacks, spoofing attacks, and tampering attacks.

Longari et al. (Longari et al., 2020) specifically designed an intrusion detection system for Internet of Vehicles named CANnol, which is adopted in an unsupervised learning mode. During its training phase, data are automatically analyzed without needing to understand the semantics of the messages; CAN messages are received, and subsequent sequences are predicted. Anomalies are then detected based on the reconstruction error between actual and predicted data. Song et al. (Song et al., 2020) introduced an intrusion detection system for the Internet of Vehicles based on deep convolutional neural networks (DCNN) and developed a new module named "frame builder" that allows direct input of CAN communication traffic into the proposed model. The results indicated that this system could effectively detect four types of attack patterns: DoS, fuzzing, spoofing of driver devices, and spoofing RPM meters. Lo et al. (Lo et al., 2022) proposed a vehicular intrusion detection system, HyDL-IDS, comprised of CNN and LSTM components. The IDS consists mainly of a CAN traffic preprocessor and a HyDL detector. The CAN traffic preprocessor first standardizes captured CAN traffic before it is input into the HyDL module. The HyDL detector, effectively the brain of the entire system, extracts spatial and temporal features from network traffic and classifies the traffic using the aforementioned deep learning methods. Javed et al. (Javed et al., 2021) combined CNN with attention-based gated recurrent units (AGRU), proposing an intrusion detection method named CANintelliIDS. The detection model, made up of three AGRU layers and two CNN layers, transforms data points into a vector sequence and inputs them into the CNN layers; these layers then extract significant features. The AGRU model learns the sequence by adjusting its weights according to the relationships and contextual information of the data points. An attention mechanism allocates weight scores based on the prominence of data point sequences, multiplies these scores with corresponding vectors, and generates context vectors to predict target labels.

Deep learning, as a data-driven representation learning technique, offers substantial advantages over traditional signature-based methods and conventional machine learning approaches for intrusion detection in the Internet of Vehicles (IoV). This includes its capability for automatic feature learning and hierarchical feature selection, which can be conducted in an unsupervised or semi-supervised manner, thereby negating the necessity for manual feature extraction. Architectures such as Artificial Neural Networks (ANN), Deep Neural Networks (DNN), Deep Belief Networks (DBN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Generative Adversarial Networks (GAN) have been extensively utilized in the domain of IoV intrusion detection (Li et al., 2021; Alladi et al., 2021; Taslimasa et al.,

2023; Korium et al., 2024). While deep learning methods can markedly enhance detection accuracy, they frequently demand significant computational and storage resources. Furthermore, the centralized model training process entails a substantial risk of privacy leakage, which is detrimental to the safeguarding of privacy information, including smart service data, vehicle locations, and user identities, within the IoV context. To date, no extant research has succeeded in developing a comprehensive IoV intrusion detection methodology that concurrently achieves high detection accuracy, low latency, lightweight implementation, and robust privacy protection.

2.2. Applications of federated learning

Since its introduction in 2016, federated learning has garnered significant attention from researchers as an effective solution to the challenges posed by isolated data (McMahan et al., 2017). This technology facilitates collaborative model training across multiple institutions without the necessity of data sharing, thereby safeguarding user privacy and security. As illustrated in Fig. 1, traditional federated learning is orchestrated by a central server. Each participating entity utilizes its local data to train a machine learning model and generates updates to the model parameters, such as weights or gradients, which are subsequently transmitted to the central server. The central server aggregates these model parameter updates from various clients, typically by computing a weighted average, and redistributes the aggregated updates back to the clients. The clients then integrate these aggregated parameters to refine their local models.

In IoT edge computing, decentralized systems and scattered data sources are well-suited for federated learning to detect malicious attacks. Fang et al. (Fang et al., 2023) proposed a federated learning-based malware detection framework, FEDroid, which leverages a genetic algorithm to simulate malware evolution and generate novel samples. Precision in detection is achieved through a custom residual neural network. FEDroid's federated learning approach allows multiple detection nodes to collaboratively train a robust detection model, thereby preventing the leakage of sensitive information. Pei et al. (Pei, 2022) introduced a semi-supervised federated learning method that integrates both labeled and unlabeled data. Central to this method is a knowledge-sharing mechanism optimized for malicious software detection processes, alongside a new network structure named Subgraph Aggregating Capsule Network (SACN), specifically designed for malware identification tasks in IoT environments. Hsu et al. (Hsu et al., 2020) proposed a privacy-preserving federated learning system for Android malware detection, named PPFL, which enables mobile devices to collectively train an optimal classifier without exposing local data.

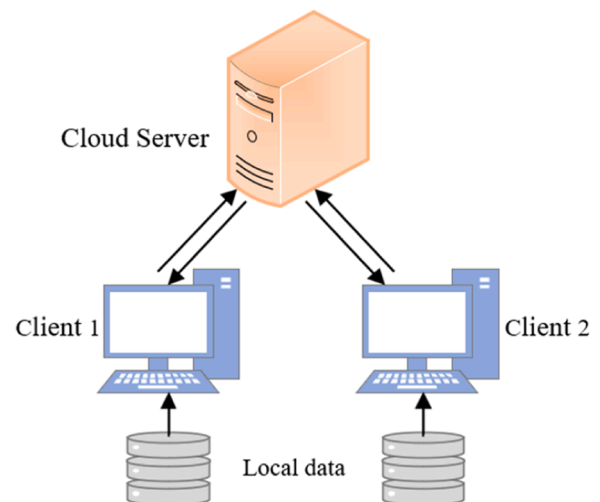


Fig. 1. Federated Learning Architecture.

Experimental results indicate that the PPFL classifier outperforms centralized training systems while maintaining the privacy of local models. However, training complex models on mobile devices with limited computational resources is impractical, and parameter communication can increase network traffic. Abdel-Basset et al. (Abdel-Basset et al., 2022) introduced a novel federated learning paradigm, FED-MAL, aimed at collaboratively training multiple distributed edge devices to detect malware. In FED-MAL, malware binary files are transformed into image format, followed by the introduction of a compact convolutional model named AM-NET to learn malware image features. Experiments conducted on publicly available malware datasets demonstrate the method's effectiveness, reliability, scalability, and generality. Additionally, there is a growing interest in privacy-preserving federated learning methods. Ling et al. (Ling et al., 2024) proposed an improved efficient heterogeneous differential privacy federated learning method that includes a secure differential privacy stochastic gradient descent protocol, targeted gradient histogram feature extraction, and calculation of each client's privacy budget weight based on noise magnitude. Yan et al. (Yan et al., 2024) constructed a practical privacy protection strategy combining cryptographic technology and network communication solutions. They employed the Kafka messaging distribution system as the hub for information exchange and applied a secure server aggregation protocol based on the Diffie-Hellman scheme, while integrating gradient differential privacy technology to defend against adversarial attacks. These methods effectively address issues of gradient leakage and adversarial attacks while maintaining training efficiency. Zhu et al. (Zhu et al., 2023) presented a novel lightweight federated learning framework that enhances the efficiency and security of model parameter updates through model compression techniques, allowing for the retention of clients' personalized settings. They developed an algorithm capable of decompressing and decrypting model parameters simultaneously. Tests have shown that this framework significantly reduces the number of parameters uploaded, thereby enhancing privacy protection effectively.

Although federated learning technology has made significant progress in the fields of malware identification and privacy protection, its application in vehicular network intrusion detection is still at an early stage. Yang et al. (Yang et al., 2022) proposed a ConvLSTM-based vehicular network intrusion detection method and a federated client selection (FCS) scheme based on proximal policy optimization (PPO) to optimize the model accuracy and system overhead of federated ConvLSTM model training, achieving detection accuracy of over 95 % on multiple datasets. However, the detection model used is a combination of convolutional neural network (CNN) and LSTM, with a complex structure unsuitable for deployment on performance-constrained in-vehicle devices. In addition, although local clients in the federated learning architecture do not need to share their private data, the method did not use an encryption scheme during model parameter aggregation. Moreover, studies have shown that the original data can be partially reconstructed through gradients (Zhu et al., 2019), or some records can be inferred to be from a specific participant based on intermediate parameters (Song et al., 2017). Therefore, the privacy protection capability of this scheme is relatively weak. Samarakoon et al. (Samarakoon et al., 2019) proposed a federated learning-assisted vehicular communication framework, where vehicles act as workers to send learned contextual results to nearby roadside units (RSUs) for network power optimization in vehicle-to-vehicle communication networks. This method meets ultra-low latency and high reliability requirements. However, the data distribution assumed by this method is a standard independent and identically distributed scenario, which does not conform to real-world conditions. Additionally, it does not consider network intrusion attacks, so its practical application capability is limited. Chai et al. (Chai et al., 2020) proposed a hierarchical blockchain framework and a hierarchical vehicular federated learning algorithm, wherein vehicles learn environmental data through machine learning methods and share the learned knowledge. This sharing process is modeled as a market

process to incentivize sharing behavior and is expressed through multi-leader and multi-follower games. Simulation results show that this algorithm can improve sharing efficiency and learning quality. However, the datasets used are MNIST and CIFAR-10, which are image datasets in the visual domain and not vehicular network security datasets. Therefore, its applicability in real vehicular network environments remains to be verified. Additionally, the communication overhead of the hierarchical federated learning framework, which is crucial for vehicular network environments with large data volumes and numerous nodes, was not discussed.

Currently, the implementation of federated learning in Internet of Vehicles faces the following challenges:

First, the diverse security scenarios and complex requirements in Internet of Vehicles make existing privacy protection methods challenging to apply universally. Traditional centralized machine learning privacy protection research assumes the central server's honesty and solely focuses on external attacks after model deployment, lacking mechanisms to address internal attacks. In addition, existing algorithms typically focus on single centralized training scenarios, often ignoring data exchange and collaborative model training under multiple participants, architectures, and data distribution patterns. Therefore, designing privacy protection algorithms that can adapt to different scenarios and requirements, and resist both external and internal attacks, is a major challenge for applying federated learning in Internet of Vehicles.

Second, due to the low trustworthiness of participants and multiple potential attack angles in federated learning, the privacy protection algorithm necessitates greater robustness. Robustness here refers to the model's ability to remain stable under malicious attacks. In federated learning, once an attacker launches an attack, they can observe and alter numerous intermediate parameters, making privacy defense much more difficult than external attacks. Colluding participants could access more sensitive information. Therefore, enhancing algorithm robustness and lessening reliance on participant trustworthiness is a critical challenge in federated learning privacy protection.

Third, federated learning itself exhibits unstable communication and significant computational cost, while the vehicular network environment imposes strict requirements on communication volume and complexity. The complex and loose network structure of Internet of Vehicles leads to unstable terminal communication, and the intricate privacy protection algorithm brings higher computational and communication costs, thus constraining the training efficiency of federated learning. Therefore, developing optimal and streamlined federated learning privacy protection algorithms to mitigate overhead remains a core challenge in vehicular network federated learning.

Fourth, the uneven distribution of vehicle traffic data makes it difficult to maintain model usability while protecting privacy. The dimension of the federated learning intermediate parameters is related to the model structure and input data dimensions, and the parameter dimension is usually extremely high, resulting in significant privacy costs. In addition, the number of users is uncertain, leading to varied dataset sizes. Balancing these varied datasets while protecting privacy is a considerable challenge.

Considering these factors, exploring a comprehensive federated learning architecture applicable to intrusion detection in Internet of Vehicles has significant research value and application prospects.

3. The proposed method

In this section, we first introduce the proposed framework for the intrusion detection method for Internet of Vehicles. Subsequently, we describe the data transformation process, where tabular vehicular traffic data is converted into three-channel images. Following that, an enhanced, lightweight deep learning model is designed specifically for the image dataset, capable of effectively recognizing images of intrusive traffic. Lastly, we elaborate on the workflow of the FED-IoV scheme,

which is based on a federated learning architecture.

3.1. Overall architecture

The federated learning-based framework for Internet of Vehicles (IoV) intrusion detection is shown in Fig. 2. This framework is composed of three components: the Trust Authority, the Cloud Server, and the Clients.

3.1.1. Trust authority

Its main function is to provide public and private keys, facilitating secure communication between clients and the cloud server throughout the training process. Hence, a secure channel for parameter updates can be established between clients and the server, ensuring the privacy of client data.

3.1.2. Cloud server

The purpose of the cloud server is to construct a global intrusion detection model by exchanging model parameters with the clients and aggregating all client parameters.

3.1.3. Client

Each client represents a local intrusion detection model. The primary responsibilities of each detection agency include local dataset collection, data preprocessing, and local model training. After completing local model training, clients interact with the server under encryption to update model parameters.

Federated learning encompasses stages including parameter uploading, distribution, transmission, and model aggregation. Although federated learning offers enhanced privacy protection compared to traditional centralized machine learning by avoiding direct data exchange, it still cannot completely eliminate the threat of information leakage. The risk of model privacy leakage arises from inherent vulnerabilities and the capabilities of sophisticated attackers. This study explores two main types of privacy attacks targeting data theft and compromising model privacy and confidentiality.

Scenario 1: Following client-side training, local parameters are uploaded to the central server. At this juncture, the central server, as a potential attacker, can utilize the collected local parameters to initiate reconstruction attacks to recover the original data of the terminals or conduct attribute inference attacks to determine if the data contains sensitive attributes.

Scenario 2: Upon aggregating terminal parameters, the central server redistributes the global parameters to each terminal. At this juncture, untrusted terminals may mount attacks, using global intermediate parameters to reconstruct samples of a certain category, infer whether a sensitive attribute exists in a particular record, and deduce the conditions of the individual owning that record from the data's common features.

To address these privacy leakage risks, this paper introduces a Trust Authority and homomorphic encryption technology within the federated learning framework. Through homomorphic encryption (HE), clients can encrypt parameters before sending them to the server, allowing the server to perform operations on the data without accessing plaintext. Endpoints then decrypt the results of the encrypted operations, ensuring consistency with plaintext operations. The encryption and decryption keys are provided by a fully trusted Trust Authority, thereby ensuring effective privacy protection. The server and clients are assumed to be semi-honest parties, strictly observing the established protocol.

3.2. Data preprocessing

A. Data Description: The CAN bus is currently the most commonly used vehicle internal communication network protocol in automobiles. It is a network based on a bus topology structure that allows ECUs (nodes) to send and receive various information relating to control of steering, speed, and more. Each node broadcasts a message to the entire system in a predefined frame format. The structure of a CAN data frame is illustrated in Fig. 3, where, from left to right, the fields are as follows: Start of Frame, Arbitration ID, Control Field, Data Field (Payload), Cyclic Redundancy Check (CRC), Acknowledgment Field (ACK), and End of Frame. Among these, the CAN Identifier (ID) and the 8-bit data field of

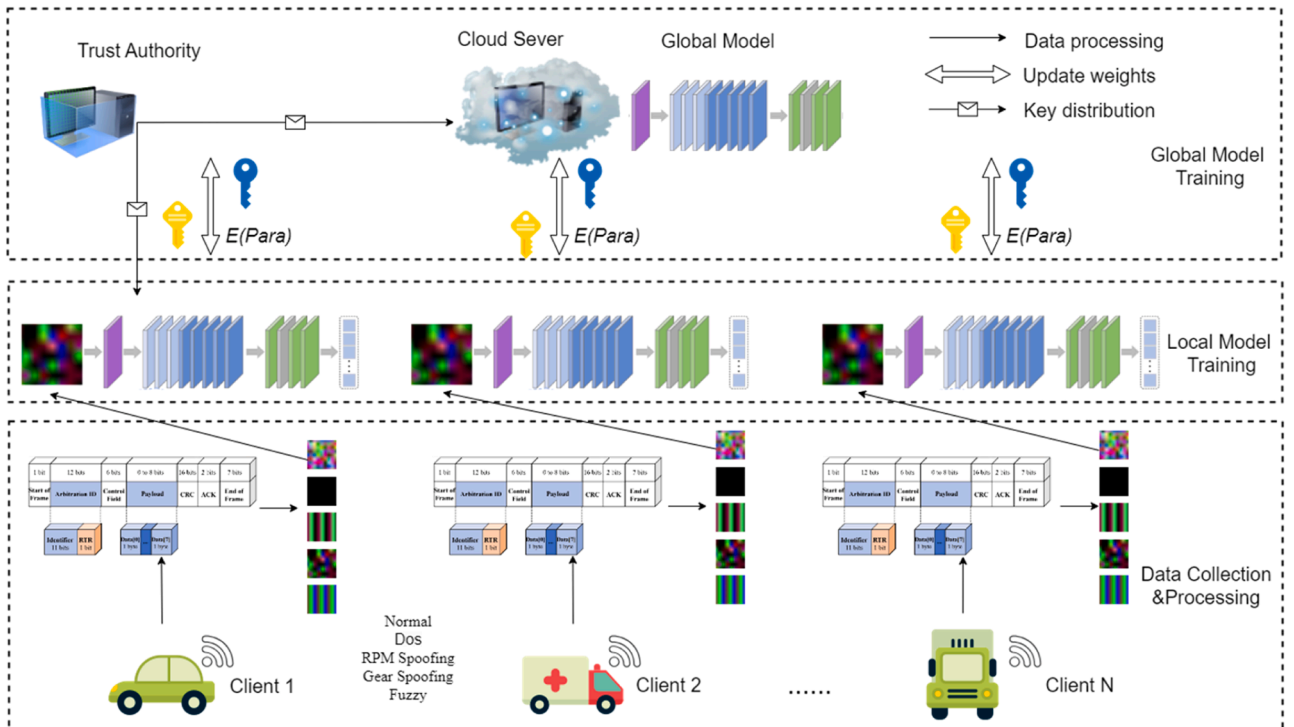


Fig. 2. Overall architecture.

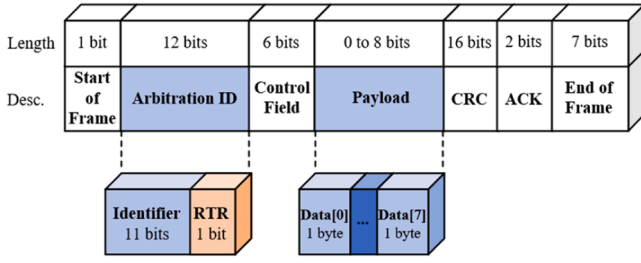


Fig. 3. CAN Frame Format.

the CAN message are the main characteristics of the communication traffic.

B. Data Transformation: During the processing of the data, it was discovered that the voluminous data stream contained "dirty data," characterized by missing data items and erroneous data item displacements. Given that data collection errors can also occur in real-world environments, it is crucial to eliminate or modify such data to prevent contamination of the dataset and ensure that training and testing results are not adversely affected. The first step involves data cleaning of the dataset, where original samples with abnormal data positioning are corrected, missing data are set to 0, and original samples with irregular data formats are deleted.

Considering that vehicular network intrusion detection must be both rapid and effective, and given the superiority of CNN models in the image domain, it is necessary to transform the intrusion detection task into an image classification problem. Hence, we transform the tabulated network traffic data into images. Initially, the original four feature data from Table 1 are split and transformed into nine features. The results of the split are illustrated in Table 2, where ID represents the CAN identifier, and the 8-bit CAN data packet is divided into eight columns (D0-D7), with each column containing two hexadecimal values. The hexadecimal values in ID and DATA are transformed into decimal values, followed by data normalization.

After data normalization, the data samples are transformed into data blocks based on the timestamps and feature sizes in the network traffic dataset. The dataset contains 9 important features (CAN ID and DATA [0]-DATA[7]), and the 9 features of 27 consecutive samples ($27 \times 9 = 243$ feature values) are transformed into images with a shape of $9 \times 9 \times 3$. Therefore, each transformed image is a square color image with three channels (red, green, blue). Since the images are generated based on the timestamps of the data samples, the temporal sequence correlation of the original network data can be preserved. The transformed images are labeled according to the patterns of attack in the data blocks, if all samples in an image are normal, then the image is labeled as "Normal." On the other hand, if an image contains attack samples, it is labeled with the attack type that has the highest proportion in that block. For example, if a DoS attack occurs in the block with the highest proportion, the corresponding image will be labeled as a DoS attack.

Furthermore, due to the relatively small dimensionality of traffic data, bicubic interpolation (Gribbon and Bailey, 2004) is utilized to

upscale images from a $9 \times 9 \times 3$ configuration to a larger $224 \times 224 \times 3$ format. This approach aims to reduce the model's risk of overfitting. In this interpolation process, the pixel value $f(x, y_1)$ for a particular data point (x, y) is estimated using the known pixel values at four surrounding points, denoted as $f(Q_{11}), f(Q_{12}), f(Q_{21}), f(Q_{22})$, which correspond to the coordinates $(x_1, y_1), (x_1, y_2), (x_2, y_1)$ and (x_2, y_2) respectively. The implementation of linear interpolation begins by calculating the pixel values $f(x, y_1)$ and $f(x, y_2)$ through one-dimensional linear interpolation along the x-axis direction, as detailed below:

$$f(x, y_1) = \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21}) \quad (1)$$

$$f(x, y_2) = \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22}) \quad (2)$$

Following this, the pixel value $f(x, y)$ for the data point (x, y) is determined using one-dimensional linear interpolation applied along the y-axis.

$$f(x, y) = \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2) \quad (3)$$

After the above mentioned data processing steps, the resultant transformed image set is generated to serve as the input for the detection model.

Fig. 4 shows representative samples from the Car-Hacking dataset (Seo et al., 2018), the CICIDS2017 dataset (Sharafaldin et al., 2018), and the CAN-TRAIN-AND-TEST dataset (Lampe and Meng, 2024). As illustrated in Fig. 4, within the Car-Hacking dataset, distinct variations in feature patterns between normal and various attack samples are discernible from Figure 4(a). The patterns of fuzzy attack images are characterized by increased randomness compared to normal imagery, whereas DoS attack samples are mainly composed of high-frequency null messages, so the images are mostly black pixels. Gear and RPM spoofing attacks involve the injection of messages and packets with specific CAN IDs to impersonate legitimate entities, therefore displaying unique feature patterns. Thus, within the same attack category, the occurrence of lines and blocks in colors such as red and blue at analogous locations across images underscores how semantic and structural information of traffic data are represented through pixel similarities and variances, establishing them as viable features for detecting malicious attacks. Notably, the tri-channel color imagery encompasses the informational content of single-channel grayscale imagery, providing redundancy where if one channel is compromised, the remaining channels can contribute to rectification and accuracy of classification outcomes. Likewise, despite the comparatively challenging task of distinguishing between image categories associated with the CICIDS2017 dataset, as featured in Figure 4(b), discernable inter-category differences are observable. According to the feature patterns shown in Figure 4(c), there are obvious visual differences between the normal samples and the attack samples.

3.3. MobileNet-Tiny

In computer vision, lightweight models have been created to address the complexity and large size of traditional models, as well as the strict requirements for their deployment. Among these, the MobileNet (Howard et al., 2019) series, encompassing the MobileNet V1, MobileNet V2, and MobileNet V3 models, stands out. The MobileNet V1 model is characterized by its assembly of depthwise separable modules; on this foundation, the MobileNet V2 model introduces the inverted residual and linear bottleneck layer modules, forming the bottleneck residual modules. Further advancing this series, the MobileNet V3 model integrates the depthwise separable convolution modules from MobileNet V1 and the bottleneck residual modules from MobileNet V2, alongside the addition of the SE attention module and the implementation of a

Table 1
Original Data.

	Timestamp	ID	DLC	Data	Label
Car-Hacking	1,479,121,434.85022	0350	8	05 28 84 66 6d	0
				00 00 a2	
	14,781,911,10.,350,41	0316	8	45 29 24 ff 29 24	1
				00 ff	
	14,781,932,58.,154,21	043f	8	01 45 60 ff 29 00	2
				00 00	
	14,781,984,08.,902,76	0000	8	00 00 00 00 00	3
				00 00 00	
	14,781,958,80.,834,81	062e	8	95 5f 01 e7 19	4
				47 2e 12	

Table 2
Data After Splitting.

ID	Data								Label
	D0	D1	D2	D3	D4	D5	D6	D7	
0.013	0.019	0.110	0.518	0.400	0.427	0	0	0.635	0
0.012	0.267	0.161	0.141	1	0.161	0.141	0	1	1
0.017	0.004	0.270	0.376	1	0.420	0	0	0	2
0	0	0	0	0	0	0	0	0	3
0.024	0.0584	0.373	0.004	0.900	0.980	0.278	0.180	0.071	4

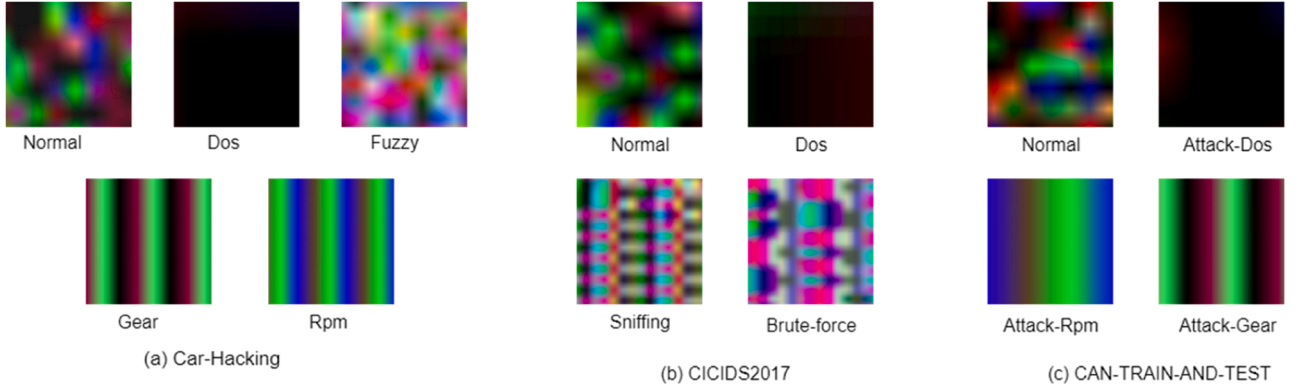


Fig. 4. Representative Samples.

novel activation function, h-swish(x). The SE module bolsters the model's capacity for learning by exploiting the relationships among channel features, and the h-swish function, known for its robust non-linear expression and progressive saturation features, is suited for integration into the convolutional and fully connected layers of deep neural networks. This supports enhanced gradient flow and optimization performance, contributing to greater model accuracy and training efficiency.

To address the relatively small feature set of the vehicular traffic data and mitigate potential overfitting from the excessive depth of the MobileNet network, we prune the original MobileNet V3 model. MobileNet V3's backbone consists of repeatedly stacked Bottleneck layers. By removing three specific Bottleneck layers (the 8th, 10th, and 12th, as shown in Table 3), we retain 13 layers from the original model. This streamlined structure better aligns with the characteristics of the vehicular network dataset, enhancing both training stability and model

reliability.

Moreover, within the MobileNet V3 architecture, the SE (Squeeze-and-Excitation) attention module focuses on encoding inter-channel relationships to determine the significance of each channel (Hu et al., 2018). This approach, however, neglects the spatial information critical for recognizing the target object's structural attributes. The transformed traffic images distinctly manifest significant variations in key pixel locations and visual textures. For instance, images associated with Gear and RPM spoofing attacks feature green vertical lines in specific regions, contrasting with the homogeneous colored block patterns observed in normal and fuzzy images. To bolster the MobileNet V3 model's capability to discern key locational features within attack samples and to enhance overall recognition accuracy, the Coordinate Attention (CA) mechanism has been integrated. Replacing the SE module within MobileNet V3 allows not only for the assimilation of cross-channel information but also the apprehension of location-sensitive details (Tsai and Tseng, 2023). This methodology emphasizes critical pixel coordinate positions, and notably, the CA module's computational footprint remains low, adhering to the prerequisites for a lightweight design.

In this study, MobileNet V-small has been adapted as the foundational architecture to accommodate the unique attributes of the vehicular network traffic image dataset. Recognizing the dataset's feature scarcity, certain layers from the original architecture have been pruned to reduce the superfluous parameter count. Moreover, aligning with the distinct visual traits of traffic imagery, the SE (Squeeze-and-Excitation) module has been supplanted by the Coordinate Attention (CA) module, thereby enhancing the model's acuity towards spatial relationships within the traffic images. These enhancements collectively contribute to reduced model complexity and susceptibility to overfitting, increased computational efficiency, and decreased computational resources and storage requirements for model deployment. The culmination of these initiatives is the creation of MobileNet-Tiny, a bespoke, lightweight model designed specifically for vehicular network intrusion detection, as depicted in Fig. 5.

Table 3
Original MobileNet V3 Structure.

Number	Input	Operation	Exp size	Out	SE	NL	Stride
1	$224^2 \times 3$	Conv2d, 3×3	—	16	—	HS	2
2	$112^2 \times 16$	Bneck, 3×3	16	16	✓	RE	2
3	$56^2 \times 16$	Bneck, 3×3	72	24	—	RE	2
4	$28^2 \times 24$	Bneck, 3×3	88	24	—	RE	1
5	$28^2 \times 24$	Bneck, 5×5	96	40	✓	HS	2
6	$14^2 \times 40$	Bneck, 5×5	240	40	✓	HS	1
7	$14^2 \times 40$	Bneck, 5×5	240	40	✓	HS	1
8	$14^2 \times 40$	Bneck, 5×5	120	48	✓	HS	1
9	$14^2 \times 48$	Bneck, 5×5	144	48	✓	HS	1
10	$14^2 \times 48$	Bneck, 5×5	288	96	✓	HS	2
11	$7^2 \times 96$	Bneck, 5×5	576	96	✓	HS	1
12	$7^2 \times 96$	Bneck, 5×5	576	96	✓	HS	1
13	$7^2 \times 96$	Conv2d, 1×1	—	576	✓	—	1
14	$7^2 \times 576$	Pool, 7×7	—	—	—	HS	1
15	$1^2 \times 576$	Conv2d, 1×1 , NBN	—	1280	—	—	1
16	12×1280	Conv2d, 1×1 , NBN	—	1000	—	—	1

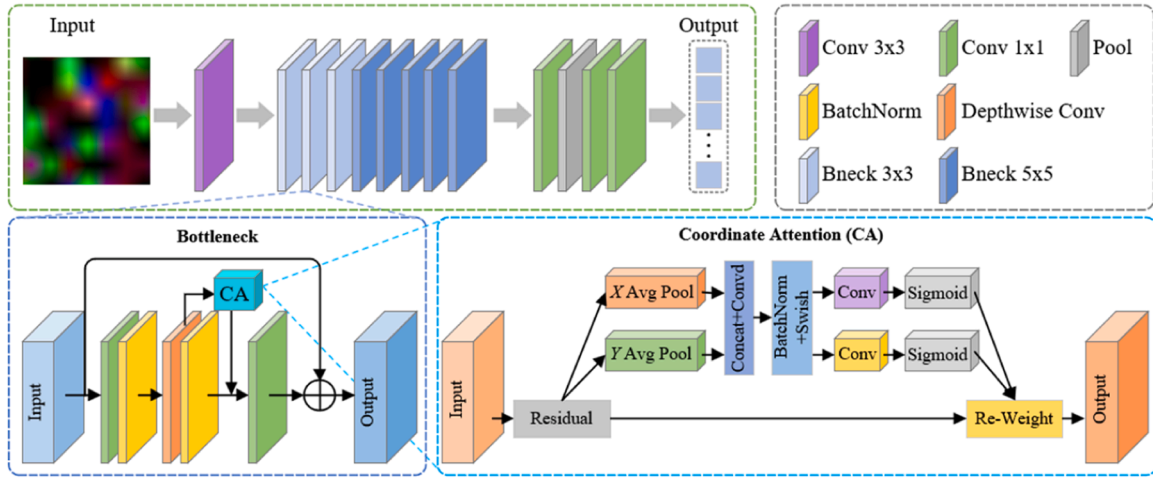


Fig. 5. Structure of MobileNet-Tiny.

3.4. Fed-IoV

FED-IoV aims to integrate multiple local intrusion detection models through a federated learning framework based on the widely used Federated Averaging algorithm, FedAvg (McMahan et al., 2017), which is designed with a client-server architecture. During training, the model structure and parameters are randomly initialized by the central server and distributed to the clients. The model is trained and updated by the clients based on local data, and the updated parameters are then uploaded to the central server. The parameters received from the clients are aggregated by the central server and the updated model parameters are distributed back to the clients. This process is repeated until the model converges or training ends.

To further protect client data privacy, a security protocol managed by a third-party trust authority is innovatively adopted by FED-IoV. An improved Paillier homomorphic encryption scheme (Hardy et al., 2017) is used by this protocol to achieve encryption of data transfer and decryption processing, utilizing a 1024-bit key and implemented through python-paillier (python-paillier 2017). Parameter aggregation without exposing the original data is allowed by this homomorphic encryption scheme, thereby ensuring data privacy is not compromised. The workflow of FED-IoV is divided into five stages:

a) System Initialization

An authoritative institution generates a key pair to ensure the security and privacy of data transmission during model training. By selecting two large prime numbers, p and q , it calculates $N = pq$, $L(x) = (x - 1)/N$, and establishes a public key $\mathcal{PK} = (N, g)$, as well as constructs a private key $\mathcal{SK} = (\lambda, \mu)$, where λ is the least common multiple of $p - 1$ and $q - 1$, and μ is the multiplicative inverse modulo N , $\mu = (L(g^{\lambda} \bmod N^2))^{-1}$. At initialization, the private key $\mathcal{SK}$ is distributed to each client, supporting secure decryption. Additionally, the authoritative institution allocates symmetric key Si to each pair of clients and servers, and assigns a unique port number to ensure precise matching in communication, thereby creating a secure communication and training infrastructure.

a) Local Model Training

Each client independently trains a deep learning model for vehicular network intrusion detection on their local vehicular edge devices. Utilizing the data transformation method described earlier, vehicular traffic data is transformed into color images, the MobileNet-Tiny model is employed to train the image dataset until convergence.

a) Parameter Encryption

After the local model training is complete, the clients encrypt the model parameters. For any parameter $w_k^r \in \mathbb{Z}_N$, with a random number $r \in \mathbb{Z}_N^*$ satisfying $r < N$ each model's parameters are encrypted using the public key $\mathcal{PK}$ before being sent to the server. The generated ciphertext $\widetilde{w}_k^r$ is computed as follows:

$$\widetilde{w}_k^r = E_{\mathcal{PK}}(w_k^r) = g^{w_k^r r^N \bmod N^2} \quad (4)$$

a) Parameter Aggregation

When the server receives the model parameters from each client, it aggregates the model parameters according to the following formula:

$$\begin{aligned} S &= \prod_{i=1}^K E_{\mathcal{PK}}(\widetilde{w}_k^r) \\ &= g^{w_k^r r_1^N \cdot g^{w_k^r r_2^N} \dots g^{w_k^r r_K^N \bmod N^2}} \\ &= g^{\sum_{i=1}^K w_k^r \cdot \prod_{i=1}^K r_i^N \bmod N^2} \end{aligned} \quad (5)$$

Then, the model parameters are calculated in the case of encryption, the formula is as follows:

$$w_{\text{avg}}^r = \frac{1}{K} \prod_{i=1}^K E_{\mathcal{PK}}(\widetilde{w}_k^r) \quad (6)$$

a) Parameter Update

After aggregating the encrypted model parameters from the clients, the server sends the updated parameters back to the clients, the clients then decrypt these parameters and update their models accordingly.

$$\begin{aligned} w_{\text{sum}}^r &= L(\widetilde{w}_{\text{avg}}^r \bmod N^2) \cdot \mu \bmod N \\ &= \frac{L(g^{\sum_{i=1}^K \widetilde{w}_{\text{avg}}^r \cdot \prod_{i=1}^K r_i^N \bmod N^2})}{L(g^{\lambda \bmod N^2})} \bmod N \\ &= \sum_{i=1}^K \widetilde{w}_{\text{avg}}^r \bmod N \end{aligned} \quad (7)$$

The interaction described above between the clients and the server for updating model parameters, after R rounds, results in obtaining a

globally optimized vehicular network intrusion detection model. The entire algorithmic process is summarized as shown in [Algorithm 1](#).

4. Experiments and analysis

This section elaborates in detail on the experimental setup of the proposed method on authoritative public datasets, incorporating comparative experiments on local model performance, comparisons with federated learning architectures, and contrasts with existing methods in the literature. Additionally, it provides an in-depth discussion of the experimental outcomes.

4.1. Experimental environment

The experiments were conducted using Python 3.7 as the programming language and Pytorch for developing deep learning models. On the hardware front, the training of deep learning models was carried out on a workstation equipped with a 64-bit Windows 10 operating system, an Intel Xeon(R) Silver 4210 CPU @ 2.20 GHz, an NVIDIA GeForce RTX 2080ti GPU (11 G), and 256GB of memory. Deployment and testing were performed on a Raspberry Pi 4 (8GB) to simulate the central server in a vehicular network and the vehicle-level local machines, respectively.

4.2. Datasets

The experiments were conducted utilizing two established Internet of Vehicles (IoV) security datasets, specifically the Car-Hacking ([Seo et al., 2018](#)) and CICIDS2017 ([Sharafaldin et al., 2018](#)) datasets. The Car-Hacking dataset, compiled in 2018, involved the use of two custom Raspberry Pi devices installed in a Kia SOUL vehicle: one for executing information intrusion attacks and the other for capturing CAN bus traffic. The aggregated CAN communication data recorded by these devices constituted the dataset. Conversely, the CICIDS2017 dataset was collected from a real-world environment and is extensively utilized in studies simulating IoV network intrusion detection. The

CAN-TRAIN-AND-TEST (CTAT) dataset ([Lampe and Meng, 2024](#)), the most recent CAN dataset released in 2024, comprises one training set and four test sets. This dataset includes CAN data from four different vehicles manufactured by two different companies, collected during real Internet of Vehicles operations. It encompasses a significant amount of normal vehicle traffic as well as nine types of attack traffic, all classified as "Attack." Details regarding the attack types and their quantities in these three datasets are presented in [Table 4](#).

4.3. Evaluation metrics

To comprehensively evaluate the classification performance of FED-IoV, multiple metrics were utilized. Initially, five-fold cross-validation was adopted, in which the dataset was partitioned into a training set (4/5) and a test set (1/5). Following each cross-validation, overall average accuracy, precision, and recall were calculated. Overall accuracy is defined as the number of correctly predicted samples divided by the total number of samples. Overall precision and recall are the weighted averages of the precision and recall of each class, considering the impact of varying sample sizes across classes on the evaluation results. True

Table 4
Types and Quantities.

	Type	Quantities	Label
Car-hacking	Normal	988,872	0
	RPM Spoofing	654,897	1
	Gear Spoofing	597,252	2
	Dos	587,521	3
	Fuzzy	491,847	4
CICIDS2017	Normal	2273,097	0
	Dos	380,699	1
	Sniffing	158,930	2
	Brute-force	13,835	3
CTAT	Normal	8011,130	0
	Attack	105,130	1

Algorithm 1

Fed-IoV.

Input: w_k^r, C, D_k, R
Output: The comprehensive Detection Model

1 Initialization
(1). The trust authority generates keys for the server and each detection authority $\{\mathcal{P}, \mathcal{K}, S\mathcal{K}\}$;
(2). The trust authority assigns port numbers to the server and detection agency to establish a secure channel;
(3). Cloud server initialization hyperparameters and model parameters w_k^r ;
(4). Initialize the communication round index by $r = 1$;

2 Procedure:
for r **in** $\text{range}(R)$ **do**:
(1). For local detection agents:
For $\forall i \in C_i, i \in [1, 2, \dots, k]$ **do**
a). C_i compute the r round local model parameters w_k^r ;
b). Encrypt model parameters by $E_{\mathcal{P}, \mathcal{K}}(w_k^r) \leftarrow \tilde{w}_k^r$;
c). C_i Upload the encrypted model parameters $\tilde{w}_k^r$ to the cloud sever;
End
(2). For cloud server:
For $\forall i \in C_i, i \in [1, 2, \dots, k]$ **do**
a). cloud server Secure Aggregation models parameters by $\prod_{i=1}^K (\tilde{w}_1^r, \tilde{w}_2^r, \dots, \tilde{w}_k^r)$;
b). The cloud server calculates the model parameters after aggregation by $w_{\text{avg}}^r \leftarrow \frac{1}{k} \prod_{i=1}^K E_{\mathcal{P}, \mathcal{K}}(\tilde{w}_k^r)$;
c). The cloud server distributes the w_{avg}^r to each client;
End
(3). For local detection agents:
For $\forall i \in C_i, i \in [1, 2, \dots, k]$ **do**
a). C_i decrypt parameters w_{avg}^r by $S\mathcal{K}$;
b). C_i updates its local model using updated parameters;
End
 $r \leftarrow r + 1$;

Return The comprehensive Detection Model

Positive (TP) denotes the number of correctly classified malicious attacks; False Positive (FP) indicates the number of benign instances misclassified as malicious; True Negative (TN) represents correctly classified benign traffic; False Negative (FN) refers to malicious attacks incorrectly classified as benign. This comprehensive evaluation method offers a more complete understanding of FED-IoV's classification performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (8)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

Moreover, given that vehicular devices typically have lower computational performance, to evaluate the efficiency of the proposed method and to facilitate a clearer comparison of performance, this study also recorded and compared training time on server-grade machines (Training Time), testing time on vehicular-grade machines (Testing Time), and the number of model parameters (Params). Training time reflects the convergence speed of the method, testing time demonstrates the real-time detection capability of the method, and the number of parameters indicates the size of the deep learning model used by the method.

4.4. Experimental results

A. Experimental Results of Model Improvement Ablation

To evaluate the enhancements made to the MobileNet V3 architecture, this study uses MobileNet V3 as the baseline. The impact of network compression is tested by removing 1 to 5 Bottleneck layers (Bneck, 5×5 , starting from the 5th row in Table 3). The models with n Bottleneck layers removed are denoted as MobileNet (- n). Additionally, MobileNet V3 has been modified by the removal of 3 Bottleneck layers and the replacement of the SE attention mechanism with the CA attention mechanism, resulting in the final local model, MobileNet-Tiny. During the experiments, samples from the Car-Hacking and CICIDS2017 datasets are converted into images according to the data conversion method described in Section 3.2, and then input into the deep learning model for local training. The specific parameter settings of the local models are presented in Table 5, and the experimental results are shown in Table 6.

It can be observed from Table 6 that on the Car-Hacking and CICIDS2017 datasets, the detection indicators improve when up to three Bottleneck layers of the original MobileNet V3 are removed. When three layers are removed, the detection accuracy is maximized, with MobileNet-Tiny exhibiting 0.05 % and 0.03 % higher detection accuracy than MobileNet V3 for the respective datasets. Although the improvement is small, it indicates that the simplified model is better suited to fitting the vehicular network dataset than the more complex model. Additionally, due to the removal of three layers, the model's parameter size is reduced to 2.22 M, which is 0.57 M less, accounting for 79.57 % of the original size, thus benefiting lightweight deployment. However, when 4 and 5 Bottleneck layers are removed, although the model's

Table 5
Local Model Parameter Settings.

Hyper-parameters	Value
Epoch	50
Batch Size	64
Optimizer	Adam
Momentum	0.9
Learning Rate	0.001

Table 6
Experimental Results of Each Model.

Dataset	Car-Hacking			CICIDS			Params (M)
	Acc (%)	P(%)	R(%)	Acc (%)	P(%)	R(%)	
MobileNet V3	98.25	98.21	98.19	97.12	97.19	97.08	2.79
MobileNet (-1)	98.25	98.24	98.21	97.13	97.22	97.08	2.60
MobileNet (-2)	98.28	98.25	98.29	97.15	97.26	97.10	2.41
MobileNet (-3)	98.30	98.27	98.32	97.15	97.28	97.16	2.22
MobileNet (-4)	97.58	97.52	96.22	96.99	96.94	96.07	2.03
MobileNet (-5)	96.12	96.05	95.59	96.71	96.80	96.01	1.84
MobileNet-Tiny	99.17	99.10	99.21	98.86	98.90	98.76	2.24

parameter size is further reduced, the detection accuracy drops sharply. Specifically, MobileNet (-4) exhibits a detection accuracy that is 0.72 % and 0.37 % lower than MobileNet (-3), indicating that the model structure becomes too simple to effectively fit the vehicular network dataset, while the parameters are only reduced by 0.19 M, providing little benefit. This demonstrates that removing three Bottleneck layers from MobileNet V3 can achieve the optimal balance between detection accuracy and model parameter size.

From the original MobileNet V3 structure in Table 3, it can be observed that layers 5 through 12 comprise repeat-stacked identical Bottleneck layers. Consequently, the removal of any one of these layers yields the same effect. The Exp size and Stride must be adjusted to ensure that the feature matrix size remains unchanged after the removal of a layer. For instance, the deletion of the 7th layer can be substituted by the deletion of the 8th layer, with the 8th layer's Exp size modified to 120 to maintain a feature matrix output size of $14^2 \times 48$. Similarly, the effect of deleting the 11th layer is identical to that of deleting the 12th layer. Ultimately, to simplify operations, the 8th, 10th, and 12th layers of the original MobileNet V3 are deleted in this paper.

Deleting three bottleneck layers from the MobileNet V3 model results in the MobileNet (-3) model. Further, replacing the SE attention mechanism with the CA attention mechanism forms the final lightweight local model—MobileNet-Tiny. Experimental results indicate that the enhanced MobileNet-Tiny exhibits significant improvements in all key performance indicators on the Car-Hacking and CICIDS2017 datasets. Specifically, the detection accuracy of the model on the Car-Hacking dataset reaches 99.17 %, an increase of 0.92 % compared to MobileNet V3; on the CICIDS2017 dataset, the accuracy is 98.86 %, an increase of 1.74 %. Simultaneously, the model's parameter size is reduced to 2.24 M, which is 0.57 M less than the original, amounting to only 80.29 % of the original model. Compared to MobileNet (-3), MobileNet-Tiny increased the parameter size by only 0.02 M but significantly enhanced detection accuracy. These results demonstrate that the model pruning method proposed in Section 3.3 successfully removed unnecessary parameter structures, thus significantly reducing resource consumption. The application of the CA attention mechanism, adapted to the characteristics of the vehicular network intrusion detection image dataset, enhanced the focus on positional differences of pixels, slightly increasing the parameter amount while effectively improving the model's detection accuracy.

B. Experimental Results of Various Local Model

The experimental setup includes evaluating the data conversion method and the MobileNet V3-Tiny model against two categories of mainstream models. The first category comprises complex deep learning models like ResNet54 and Vision Transformer (ViT), while the second

category includes lightweight models such as EfficientNet-B0, ShuffleNet V2, and MobileNet V3. The evaluation focuses on both accuracy and real-time performance. The results of these comparative experiments are detailed in Tables 7 and 8.

In Tables 7 and 8, Acc, P, R, and Testing Time refer to the accuracy, precision, recall, and a single sample testing time, respectively, as obtained from experiments conducted on the Raspberry Pi device. Training time is the time it takes to train the model to converge on a server machine with a GPU. Params refers to the total number of trainable parameters in the model, which is directly related to computational complexity and memory usage.

The data from Tables 7 and 8 show that all classification models have achieved a detection accuracy of over 96 % on the Car-Hacking dataset and over 95 % on the CICIDS2017 dataset, confirming the effectiveness of the data transformation method described in Section 3.2. This is because, after transforming the traffic data into three-channel images, the intrinsic features and structural characteristics of the traffic are transformed into texture features and visual characteristics of the images. Deep learning models are capable of effectively distinguishing between features of different categories of images. Moreover, with the presence of timestamps, the models are able to mine byte sequence features and deeper characteristics. Additionally, the features of the three channels complement each other, enabling better correction of the detection results.

In comparison, the ResNet54 model, although slightly superior in detection accuracy, does not show a particularly significant advantage. On the two datasets, it only surpasses MobileNet-Tiny by 0.03 % and 0.08 %, respectively. However, on server-grade machines, the total training time for ResNet54 is 3.36 times and 3.29 times that of MobileNet-Tiny. On Raspberry Pi devices, the testing time for a single sample is 16.33 times and 14.43 times longer than MobileNet-Tiny, respectively, while the model's parameter count is as much as 11.69 times larger than that of MobileNet-Tiny. Similarly, while Vision Transformer (ViT)'s can be successfully trained on a server machine, deploying it for testing on a Raspberry Pi device results in a memory overflow that fails to predict samples. The primary reasons for these significant differences are the deeper and more complex network structures of ResNet54 and ViT, which give them stronger model fitting capabilities. Yet, their large model sizes also lead to longer training and testing times, as well as higher hardware resource consumption, making them unsuitable for edge devices with limited computing capabilities.

In the Internet of Vehicles (IoV) intrusion detection systems, the high demand for real-time performance typically requires that the detection time for each data packet be <10 milliseconds (Moubayed et al., 2020). Experimental results show that only ShuffleNet V2 (achieving 8 ms and 9 ms, respectively) and MobileNet-Tiny (at 6 ms and 7 ms, respectively) fulfill this requirement. It's noteworthy that, in comparison to ShuffleNet V2, MobileNet-Tiny exhibits higher detection accuracies by 3.13 % and 3.45 %, respectively, while its number of parameters is only 84.5 % of ShuffleNet V2's. Considering both detection performance and computational efficiency, the enhanced MobileNet-Tiny model

Table 7
Experimental Results on the Car-Hacking Dataset.

Model	Acc (%)	P(%)	R(%)	Training Time(s)	Testing Time(ms)	Params (M)
ResNet54	99.20	99.15	99.21	10,462	98	26.18
ViT	–	–	–	32,834	Out of memory	85.76
EfficientNet-B0	97.46	97.47	97.46	3865	15	5.41
ShuffleNet V2	96.04	96.10	96.02	3663	8	2.65
MobileNet V3	98.25	98.21	98.19	3491	11	2.79
MobileNet-Tiny	99.17	99.10	99.21	3137	6	2.24

Table 8
Experimental Results on the CICIDS2017 Dataset.

Model	Acc (%)	P(%)	R(%)	Training Time(s)	Testing Time(ms)	Params (M)
ResNet54	98.94	98.98	98.74	9877	101	26.18
ViT	–	–	–	30,886	Out of memory	85.76
EfficientNet-B0	95.33	94.91	95.23	3654	16	5.41
ShuffleNet V2	95.41	95.56	95.73	3468	9	2.65
MobileNet V3	97.12	97.19	97.08	3428	10	2.79
MobileNet-Tiny	98.86	98.90	98.76	3006	7	2.24

demonstrated in this study showcases the best overall detection capability in the context of the IoV.

C. Experimental Results of Various Federated Learning Frameworks

To verify the detection performance of FED-IoV in a federated scenario, we compared three existing federated learning detection methods. The Car-Hacking and CICIDS2017 datasets were randomly and evenly divided into K client datasets, with the total number of samples consistent across clients but the number of samples in each class varying, in order to simulate the uneven distribution of attacks received by each in-vehicle node in the real world. We reproduced DeepFed (CNN-GRU) (Li et al., 2020), FedCV (MobileNet V3) (He et al., 2021), and ViT-FL (Vision Transformer) (Qu et al., 2022) and compared them with the proposed FED-IoV (MobileNet V3-Tiny). To ensure data reliability, we conducted multiple repeated experiments under different settings, with the number of clients K set to (3, 5, 7) and the communication rounds R set to 10. The FED-IoV training process is as described in Section 3.4, and the local model parameter settings are shown in Table 5. The detection accuracy of each method is shown in Table 9, and the communication overhead (calculated as the number of parameters exchanged between clients and the cloud server, with $K = 3$ and $R = 10$) is shown in Fig. 6.

From Table 9, it is observed that on the Car-Hacking dataset, FED-IoV (MobileNet-Tiny) has an average detection accuracy of 98.51 %, which surpasses DeepFed (CNN-GRU), FedCV (MobileNet V3), and ViT-FL (Vision Transformer) by 6.89 %, 1.87 %, and 4.23 % respectively. On the CICIDS2017 dataset, FED-IoV (MobileNet-Tiny) achieves an average detection accuracy of 97.74 %, outperforming DeepFed (CNN-GRU), FedCV (MobileNet V3), and ViT-FL (Vision Transformer) by 6.61 %, 3.70 %, and 5.36 % respectively. As shown in Fig. 7, in terms of communication overhead, FED-IoV incurs significantly less, at 165 MB and 184 MB respectively, compared to DeepFed (1250 MB, 1366 MB), FedCV (778 MB, 820 MB), and ViT-FL (2483 MB, 2489 MB). These results indicate that, under test conditions with different numbers of federated clients ($K = 3, 5, 7$), the FED-IoV method proposed in this study significantly outperforms the existing four federated learning detection techniques in both detection accuracy and communication overhead. This is attributed to the lightweight advantage and efficient

Table 9
Results on Various Federated Learning Frameworks.

K	Datasets	DeepFed	FedCV	ViT-FL	FED-IoV
3	Car-Hacking	91.84	96.64	94.34	98.55
	CICIDS2017	91.07	94.12	91.16	97.83
5	Car-Hacking	92.46	97.28	95.51	99.07
	CICIDS2017	91.89	94.95	93.20	98.28
7	Car-Hacking	90.66	96.01	92.99	97.92
	CICIDS2017	90.45	93.06	92.78	97.11
Ave	Car-Hacking	91.65	96.64	94.28	98.51
	CICIDS2017	91.13	94.04	92.38	97.74

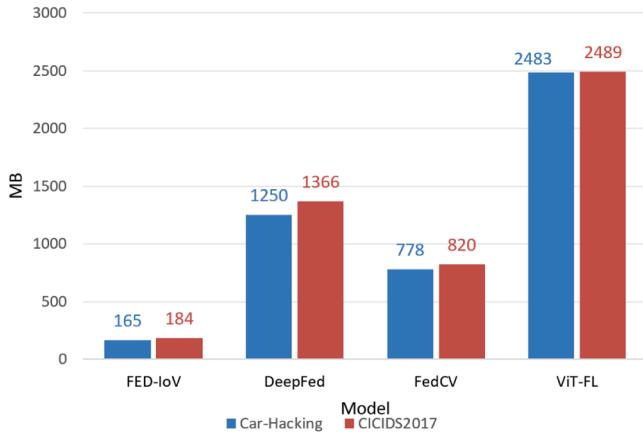


Fig. 6. Communication Overhead.

communication encryption and decryption of MobileNet-Tiny within FED-IoV, validating that FED-IoV better meets the high accuracy, deployability, and privacy protection requirements in IoV intrusion detection tasks under federated environments.

To further explore the real-world application potential of the FED-IoV method, the complexity of joint training also increases as the number of clients gradually grows, which is commonly used to test the scalability of the model. In this experiment, we increased the number of clients from 3 to 29, set the communication rounds to $R = 10$, and trained the FED-IoV model at each level. As shown in the experimental results in Fig. 7, as the number of clients increases, the classification accuracy of FED-IoV first improves, reaches a peak, and then gradually declines, specifically peaking when the number of clients is 5. Subsequently, as the number of clients further increases, the classification accuracy slightly declines before eventually stabilizing.

This performance decrease is due to the fact that, with the total number of data samples remaining unchanged, the amount of data available for training at each client decreases as the number of participating clients increases. Nevertheless, FED-IoV still maintains a detection accuracy of over 89 % in a multi-client setting, validating its outstanding scalability and stability. Moreover, as the scale of the Internet of Vehicles expands, the daily accumulation of vehicular data will continuously increase, which theoretically, could further enhance the classification performance of FED-IoV.

D. Experimental Results of Detection Performance in Complex Scenarios

In real Internet of Vehicles scenarios, vehicles may travel for miles and hours without being attacked. Therefore, IoV intrusion detection is a rare event prediction problem, where the rare event is vehicle attacks, and the problem is highly imbalanced. To assess the performance of FED-IoV in real IoV scenarios as accurately as possible, a portion of the CTAT dataset is utilized, divided into Normal and Attack categories, with a ratio of normal traffic to attack traffic (normal: attack) of 76:1. At the same time, the CTAT dataset is pre-divided into five subsets, including one training set and four validation sets, to explore the impact of unknown samples and different vehicle models. The training set contains 4005,630 normal samples and 52,530 attack samples, while the four test sets each contain 1001,375 normal samples and 13,150 attack samples.

1. test_01_known_vehicle_known_attack: Tests the model on known vehicles (seen in training) and known attacks (seen in training).
2. test_02_unknown_vehicle_known_attack: Tests the model on unknown vehicles (not seen in training) and known attacks (seen in training).
3. test_03_known_vehicle_unknown_attack: Tests the model on known vehicles (seen in training) and unknown attacks (not seen in training).
4. test_04_unknown_vehicle_unknown_attack: Tests the model on unknown vehicles (not seen in training) and unknown attacks (not seen in training).

Set the FED-IoV with different numbers of clients $K = 5$, communication rounds $R = 10$, and local model parameters as shown in Table 5. The experimental results are shown in Fig. 8.

According to Fig. 8, the detection accuracy of FED-IoV for both normal traffic and attack traffic exceeds 99 % across the four different test sets, suggesting a minimal impact of imbalanced data. The presence of a large number of normal traffic samples enables FED-IoV to effectively learn and distinguish the features of normal and attack traffic, aiding in accurate binary classification.

The detection accuracy is highest on the test_01_known_vehicle_known_attack set, as the vehicle types and attack types in this set were familiar during training. The detection accuracy is lowest on the test_04_unknown_vehicle_unknown_attack set, as the vehicle types and attack types in this set lacked presence in the training data, leading to weaker model generalization. Additionally, the detection accuracy on the test_02_unknown_vehicle_known_attack set is lower than that on the test_03_known_vehicle_unknown_attack set, implying that unknown vehicle types may have a greater impact on FED-IoV than unknown attack types. Therefore, training FED-IoV on varied vehicle types (e.g., types, manufacturers, models) appears to be particularly important.

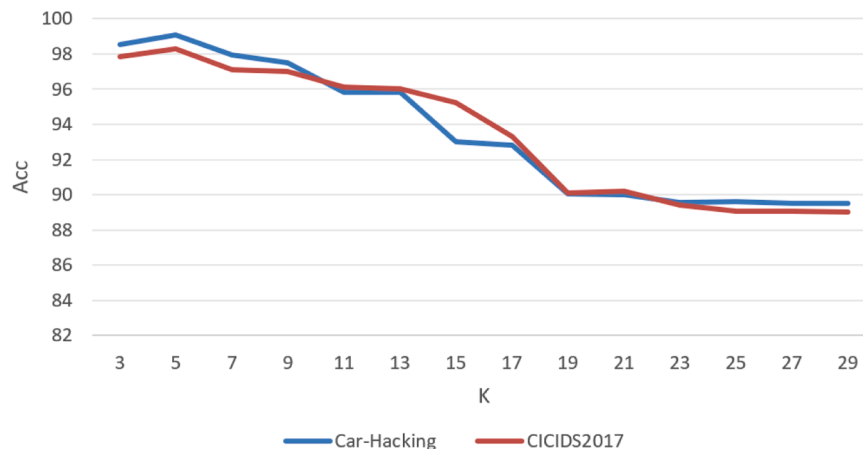


Fig. 7. Analysis of Scalability.

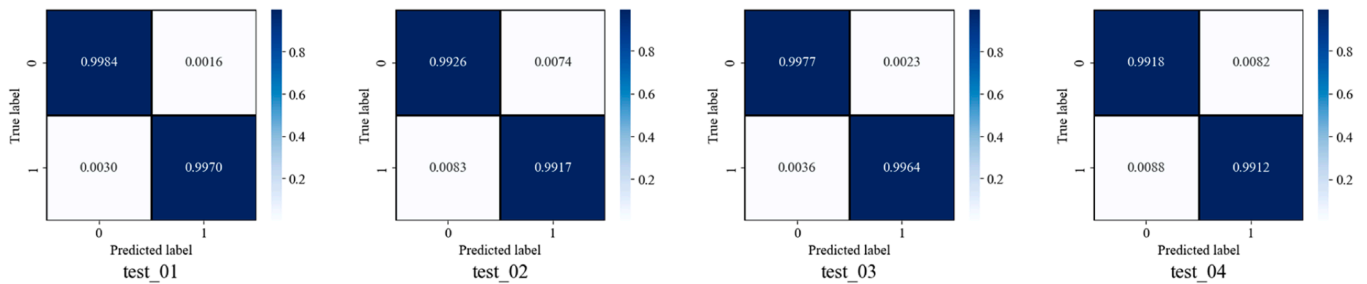


Fig. 8. Confusion matrix of experimental results on the CTAT dataset.

E. Experimental Results in Existing Literature

FED-IoV possesses high detection accuracy, can be deployed on low-computing power devices, and meets the requirements for real-time detection. Thanks to the adoption of the federated learning framework described in this paper, data from each client is trained locally, thereby eliminating the need to share personal privacy data. Furthermore, parameter updates are carried out in an encrypted manner, endowing the FED-IoV detection method with robust privacy protection characteristics. We will now compare four authoritative Internet of Vehicles (IoV) intrusion detection methods from three critical aspects: detection accuracy, real-time performance, and privacy protection.

As indicated by Table 10, the MTH-IDS method, utilizing the same Car-Hacking and CICIDS2017 datasets as this study, has achieved remarkably high detection accuracies of 99.99 % and 99.88 %. This is attributed to its reliance on manually extracted features and the establishment of a feature library, alongside the implementation of a complex dual-layer detection mechanism to further enhance accuracy. Such an approach, however, neglects the low-performance characteristics of onboard devices. Though it theoretically possesses robust detection capabilities, it is difficult to deploy in actual Vehicle Network environments, and fails to address the protection of user data privacy, thus struggling to manage the escalating issue of privacy breaches. Similarly, the GRU system and CANtelliIDS methods have employed various feature extraction techniques and deployed multiple deep learning models for detection, achieving commendable detection accuracies. Yet, the discussions in these studies have been narrowly focused on improving detection accuracy without exploring practical deployment experiments and privacy protection schemes in real-world scenarios. The LightGBM+HBGB method demonstrates a detection accuracy of 96.97 %, and it is reported in the study that it maintains real-time detection capabilities of under 10 ms even when deployed on low-performance devices. However, this approach uses a centralized training method, evidently not meeting the requirements for user data privacy protection, and its accuracy yet requires improvement.

In summary, the local detection method proposed in this paper, titled MobileNet-Tiny, benefits from a lightweight model and efficient data

transformation methods, achieving high detection accuracies of 99.17 % and 99.86 %. It simulates the scenario of low-performance onboard devices, ensuring real-time detection capability when deployed on computationally constrained devices like the Raspberry Pi. To further meet the privacy protection needs of vehicle owners in real Vehicle Network scenarios, the study explores the use of a federated learning framework for distributed training of MobileNet-Tiny, leading to the proposition of the FED-IoV detection method. This method, albeit slightly reducing detection accuracy, introduces robust privacy protection capabilities, making its overall detection ability superior to existing methods presented in the literature.

5. Conclusion

In this paper, we propose a comprehensive Internet of Vehicles (IoV) intrusion detection method named FED-IoV. Initially, based on the characteristics of vehicular traffic CAN data, we transform CAN packet DATA into three-channel color images in chronological order using data transformation techniques. Then, considering the low-performance reality of vehicular devices, we design a customized lightweight model, MobileNet-Tiny, to train and learn from the image dataset. Following this, we implement an IoV intrusion detection framework based on a federated learning architecture, which allows client models to be trained locally on their data without the need for sharing, and employs strict encryption for parameter transmission between the clients and the server, ensuring user privacy. Ultimately, on three authoritative datasets, our local model MobileNet-Tiny outperforms other deep learning models in terms of detection accuracy, real-time performance, and parameter size, making FED-IoV a comprehensive detection framework that offers high accuracy, real-time functionality, and privacy protection, satisfying the intrusion detection needs in actual IoV scenarios.

In the model parameter aggregation process, we assume that all clients are honest and only use real data for training, and the server does not maliciously modify the parameters. Therefore, the implementation of the FED-IoV framework currently has strict conditions and cannot effectively defend against data poisoning attacks. In the future, we plan to improve the comprehensive performance of the framework in the following three aspects:

1. Explore more robust federated learning methods. Study model aggregation algorithms that can still work effectively in the presence of dishonest clients. Especially focus on methods based on redundancy techniques and default detection to reduce the strict assumptions about client behavior.
2. Adopt advanced defense mechanisms to detect and resist data poisoning attacks by introducing anomaly detection. Use adaptive threshold-based malicious behavior recognition strategies to monitor and filter out anomalous data in real time. Simultaneously, explore blockchain-based distributed trust mechanisms to enhance system transparency and security by recording and verifying each client's behavior and contributions.
3. For the problem of data heterogeneity due to different vehicle devices and environments, research more adaptive model training and

Table 10
Comparison of Existing Literature Methods.

Method	Datasets	Accuracy	Real-time Performance	Privacy protection
LightGBM+HBGB (Seth et al., 2021)	CSE-CIC-IDS-2018	96.97	✓	×
GRU system (Assis et al., 2021)	CIC-DDoS-2019	97.10	×	×
CANtelliIDS (Javed et al., 2021)	CICIDS2017	94.23	×	×
MTH-IDS (Yang et al., 2021)	Car-Hacking	99.99	×	×
	CICIDS2017	99.88	×	×
MobileNet-Tiny	Car-Hacking	99.17	✓	×
	CICIDS2017	98.86	✓	×
FED-IoV	Car-Hacking	98.51	✓	✓
	CICIDS2017	97.74	✓	✓

optimization strategies. Include personalized federated learning methods that allow each client to dynamically adjust the model based on their data characteristics. Utilize transfer learning and multi-task learning techniques to address data distribution differences and improve model generalization in different scenarios.

CRedit authorship contribution statement

Kun Huang: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology. **Rundong Xian:** Data curation, Formal analysis, Methodology. **Ming Xian:** Writing – review & editing, Supervision. **Huimei Wang:** Writing – review & editing, Supervision, Software, Resources. **Lin Ni:** Project administration, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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